Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **automation-oriented software**, C++/Python systems, APIs, test pipelines, documentation, debugging workflows, and reliable deployment tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform automation software** with microservice architecture, executable builds, VM testing, and 20+ unit and integration suites using Jenkins, PyTest, Mockito, Agile reviews, release documentation, and validation.
- Troubleshot **large-scale API and ETL components** using OOP patterns, Python, SQL, AWS Glue, and S3, improving response time by **30%** across validation, deployment, release workflows, and fixes.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **modular operational services** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, integrations, monitoring, debugging, validation, releases, process models, tools, and reports.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, shared-environment handoffs, implementation support, and checks for engineers.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **database-backed software tools** for 500+ users, translating requirements into APIs, normalized schemas, monitored jobs, technical documentation, validation logic, and production releases across cloud storage systems reliably.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting production handoffs and reviews.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **concurrent backend services** handling 10,000+ daily requests, implementing secure file ingestion, automated checks, documented fixes, sprint delivery, and stable API behavior during load testing cycles.
- Partnered with users to debug **workflow and execution defects** by reproducing OCR, persistence, and processing failures, documenting regression cases, reviewing fixes, and cutting feedback turnaround by **60%**.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Analyzed **distributed consensus mechanisms** and developed a novel algorithm reducing centralized-control and quantum-risk exposure by **25%** in benchmark simulations with technical documentation, diagrams, evaluation notes, and summaries.
- Contributed to **peer-reviewed technical research** by integrating 5+ lattice-based cryptographic primitives into protocol design and presenting implementation tradeoffs, architecture decisions, reliability considerations, and results at IISc Bengaluru.

Skills

Languages:	C++, Python, C#/.NET familiarity, Java, SQL, JavaScript, TypeScript, Bash, HTML/CSS
Systems:	OOP, software design, debugging, validation, technical documentation, process models, operating systems
Automation:	CI/CD, Jenkins, Docker, VM testing, executable builds, hardware-integration interest, robotics/sensor readiness
Data:	PostgreSQL, MySQL, SQL Server, MongoDB, AWS Glue/S3, ETL, indexing, performance tuning
Delivery:	Git, GitHub, code reviews, Agile/Scrum, PyTest, Mockito, release support, troubleshooting, production handoffs

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript, OS Algorithms*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, React controls, Gantt visualization, dynamic timing calculations, debugging workflows, tests, and metrics across simulated processes.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting and automated tests reliably.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, technical reviews, and validation.